

Field Intelligence

AI-assisted field audit POC - approach, measured results and next steps

Approach

I treated this as a trust system, not a checklist demo. In a franchise network, the expensive error is a confident finding the evidence does not support: it wastes the consultant's time and damages the franchisor-franchisee relationship. The product therefore follows one rule: **the agent proposes; a human decides**.

The primary in-the-moment user is the field consultant. The mobile-first Field Companion guides them area by area while they speak, type, take a photo, record a short video, or answer structured checks. Voice is transcribed but cannot enter analysis until the consultant confirms it; photo/video models can describe observable facts but their schemas cannot return a violation. Visually verifiable or high-risk issues require a photo targeted to that exact report; other issues get an AI recommendation and may continue with detailed text at lower confidence. The agent then investigates through tenant-scoped, read-only tools and makes a schema-validated decision. Ambiguous input such as "the restroom floor looked a little dirty" becomes a targeted question, never a finding. Specific input can become a candidate finding containing the consultant's exact statement, the model's separate interpretation, cited standard, severity, confidence, uncertainty, what the evidence does not establish, and a proposed owner/deadline.

Four gates protect the reviewer: deterministic ambiguity policy, proof that the cited standard was retrieved in that run, a check that customer sentiment is not presented as field evidence, and three independent challenge lenses (evidence sufficiency, franchisee advocate, standards fit). The panel is reviewer-triggered in the SQLite POC so a consultant is not blocked for ~40 seconds in the field; missing challenger responses fail closed. Only an independent human approval creates a corrective action. Closure requires a real after-photo before manager verification; review, edits, disputes and verification are append-only events.

The implementation is a modular monolith: FastAPI, SQLAlchemy, React/Vite and one model gateway supporting Gemini or a labelled deterministic fixture engine. Single-worker SQLite keeps the POC portable; Postgres and object storage remain an explicit production migration. Prompts are versioned files and every model call records provider, model, tokens, latency, estimated cost, retries and success. A second EV/depot tenant runs through the same engine with separate standards, demonstrating the multi-tenant data boundary. A signed shared demo login protects the public POC without claiming production user-level authentication or RBAC.

Results

Google Places worked but returned five Google-selected reviews, all positive in the observed response. I kept that source visible and correctly labelled, then used a pinned open-source collector once, outside the application request path, to obtain the complete Wolf Creek public snapshot. The import removes reviewer names, profiles and photos and hashes review IDs. The resulting 362 rows include 42 one-star, 17 two-star, 28 three-star, 84 four-star and 191 five-star reviews. The product filters locally to ≤ 92 days, ≤ 3 stars and written feedback: 22 recent ratings become 7 actionable written reviews. Four recurring themes emerge: hydration availability, service/value response, cart/GPS reliability, and temporary-green disclosure. Reviews remain context and can never create a compliance finding.

I extended that signal into an operational loop: recurring themes create idempotent, assigned triage tickets; staff must validate the issue on site, attach before and after images, submit a resolution, and obtain independent manager verification. Only then does the app draft a public owner reply. It does not pretend Google exposes private reviewer contact details. Rating impact stays BASELINE_ONLY until a later comparable snapshot exists, so the product cannot manufacture an ROI claim.

"Continuous learning" is also governed. Repeated customer language can propose a new measurable parameter with anonymized examples, but a named standards owner must approve or reject it. Approval queues design work; it does not silently retrain a model or rewrite a standard.

For competitive intelligence, I collected and privacy-minimized 1,235 additional reviews from three nearby Atlanta public courses. Only aggregate counts, rates and hashed evidence references ship. Positive-theme rates show Wolf Creek's relative strengths in course condition, staff hospitality and layout/challenge, and supported opportunities in practice-facility visibility and value messaging. The UI labels the manually selected cohort directional, not representative market research.

Validation is deliberately harder than a happy-path demo. Fifty-one API/domain regression tests pass, including authentication, forged provenance, cross-issue photo reuse, concurrent checklist saves, atomic model budgets, unconfirmed voice, high-privacy media, self-review and evidence-free closure. The deterministic behavioural suite ran each executable case three times: 15/15 executable cases passed, zero flaky, with one live-vision-only case explicitly skipped rather than counted green; the zero-unsupported-finding release gate cleared over a non-empty finding set. Separately, live Gemini passed the text injection case once and the vision injection case three consecutive times. The production bundle type-checks, npm audit reports zero vulnerabilities, the Docker image builds, and a credential-protected production container passed health, anonymous-denial, secure-session and static-bundle smoke checks. Desktop and 390px mobile browser validation found no console errors or horizontal overflow. Live Gemini rejected a deliberately unrelated checklist photo instead of linking it as proof, while the text-only path visibly capped confidence. Live Gemini also transcribed a real WAV, described a real MP4 with time-coded facts, refused to store that people-filled clip in a high-privacy restroom, and downgraded a candidate finding after two of three independent challenge lenses weakened it.

What is honest, and what comes next

BroadPeak supplied no controlled internal standards, credentials or private data. Wolf Creek uses a sourced, jurisdiction-labelled external guide (law, conditional requirements, industry BMP and venue policy remain distinct); representative operating prompts and the EV tenant remain labelled. The shared demo credential uses an HttpOnly signed session, but role selection still says "demo persona" and does not provide real separation of duty. Google Business Profile publishing is a draft-only integration boundary. Live scraping is not a production dependency. Image files are content-validated locally, but production still needs identity, object storage, malware scanning, retention/deletion policy and legal review.

First 30 days: load one redacted standards set, replace the shared credential with SSO/RBAC, and add authorized Business Profile access, and agree the photo/privacy policy. Days 30-60: run one property in shadow mode and measure audit time, clarification rate, unsupported findings, successful disputes and challenge-panel value. Days 60-90: if those metrics earn trust, enable one live property, compare the post-resolution review snapshot without claiming causality, and test a second real tenant. The north-star metric is not findings produced; it is findings a franchisee can successfully dispute.